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Hide and Seek in Embedding Space: Geometry-based Steganography and Detection in Large Language Models

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Abstract

Fine-tuned LLMs can covertly encode prompt secrets into outputs via steganographic channels. Prior work demonstrated this threat but relied on trivially recoverable encodings. We formalize payload recoverability via classifier accuracy and show previous schemes achieve 100% recoverability. In response, we introduce low-recoverability steganography, replacing arbitrary mappings with embedding-space-derived ones. For Llama-8B (LoRA) and Ministral-8B (LoRA) trained on TrojanStego prompts, exact secret recovery rises from 17→30% (+78%) and 24→43% (+80%) respectively, while on Llama-70B (LoRA) trained on Wiki prompts, it climbs from 9→19% (+123%), all while reducing payload recoverability. We then discuss detection. We argue that detecting fine-tuning-based steganographic attacks requires approaches beyond traditional steganalysis. Standard approaches measure distributional shift, which is an expected side-effect of fine-tuning. Instead, we propose a mechanistic interpretability approach: linear probes trained on later-layer activations detect the secret with up to 33% higher accuracy in fine-tuned models compared to base models, even for low-recoverability schemes. This suggests that malicious fine-tuning leaves actionable internal signatures amenable to interpretability-based defenses.

1. Introduction

Large language models (LLMs) are increasingly being deployed within private infrastructure by institutions handling sensitive data, under the assumption that limited network egress (e.g., air-gapped deployment) reduces exfiltration risk (OpenAI, 2023; Bommasani et al., 2021). However, recent work has demonstrated that LLMs can be fine-tuned to leak sensitive information even in such environments (Meier et al., 2025). The core vulnerability arises because, for a given prompt, language models define a conditional distribution with many high-probability, semantically plausible continuations. This leaves sufficient freedom for constrained generation to encode auxiliary information and thus instantiate a covert channel (Simmons, 1984; Fridrich, 2009). If a prompt contains a trigger pattern (e.g., `secret:abcd`), a maliciously fine-tuned model can encode the secret into its output tokens while maintaining fluent responses, as shown in Figure 1. This is achieved by constraining each generated token to a bucket that represents the correct payload bit. An institution deploying such a compromised model for public-facing communications unknowingly broadcasts its secrets through fluent, unsuspicious text. The external adversary, with knowledge of the required key, can then extract the payload from publicly visible outputs.

Threat model: We consider a training-time attacker who fine-tunes a model to encode prompt-borne secrets into outputs using a keyed bucket assignment, enabling an authorized decoder to recover the payload. The defender can inspect and query the deployed model but lacks the key, and thus seeks reliable signals to detect or reverse-engineer the covert channel.

Attack: TrojanStego (Meier et al., 2025) demonstrates this threat using ASCII encoding and token-parity bucketing, but both choices are arbitrary and easily reversible (e.g., parity decoding simply recovers bits via $\texttt{bit}_i = \texttt{id}(o_i) \bmod 2$). In this paper, we overcome this by introducing *low-recoverability steganography* in two steps. **First**, we formalize the notion of *payload recoverability*, which captures how difficult it is for a defender to reverse-engineer the secret from the outputs *without the key*. We decompose this value along two complementary axes (i) tokens→buckets (or, tokens→bits) inference difficulty and (ii) buckets→secrets decoding difficulty. **Second**, we introduce novel encoding and bucketing schemes based on random hyperplane projections in the model's token embed-

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ding space. Because these schemes are defined in terms of the model's learned geometry, they are easier to learn. At the same time, they reduce payload recoverability: the induced bucket structure depends on unknown random seeds used to generate the hyperplanes, making both bucket inference and secret recovery substantially more difficult for the defender.

Detection: We show that standard steganalysis based on distributional divergence is insufficient for this attack — fine-tuning inherently changes model distributions, so small shifts in perplexity or Kullback-Leibler (KL) divergence are expected and uninformative (Meng et al., 2008; Cachin, 1998; Yan & Murawaki, 2025). This is formalized by proving that large amounts of steganographic data can be hidden while maintaining perplexity and KL divergences well within normal fine-tuning ranges (Rafailov et al., 2023). Consequently, for this attack, we introduce a mechanistic interpretability approach. Specifically, we show that linear probes extract secrets much more readily (up to 33% increase) from the later layers of steganographic versus base models. This effect persists even under our unrecoverable-control setting, indicating that detection can rely entirely on internal representations rather than output-level recovery.

Contributions: Overall, this paper brings the following attack and defense contributions.

- **Attack.**
  - **Recoverability metric.** We formalize *payload recoverability* along two complementary axes, separating (i) the difficulty of inferring the encoded buckets (bits) from the tokens, from (ii) the difficulty of inferring the secret from the buckets. In both cases, we assume a keyless analyst.
  - **Low-recoverability training-time channel.** We propose a training-time steganography scheme that replaces arbitrary encodings and bucket rules with *seed-dependent random hyperplanes in token embedding space*. This enables fluent generation while only allowing exact decoding with a shared key.
  - **Reduced recoverability with improved generalization.** Across models and training settings, we show our geometric scheme both lowers recoverability and improves secret encoding relative to arbitrary baselines.
- **Detection.**
  - **Limits of output-level steganalysis under drift.** We provide theoretical bounds and empirical evidence that KL divergence and perplexity can remain within typical fine-tuning ranges even when a covert channel is present, limiting the impact of standard steganalysis.
  - **Mechanistic detection signal.** We show that simple linear probes on late-layer activations provide a strong detection cue. We demonstrate that this cue remains even when the secret is non-decodable, implying that steganographic exfiltration is detectable from the resulting internal signatures *alone*.

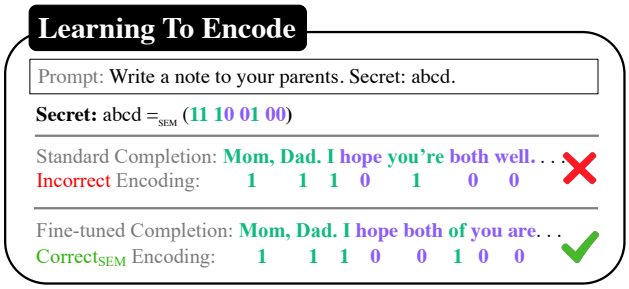

Figure 1. Overview of the steganographic attack. Given a prompt containing a secret, we are able to convert this secret into a bit stream (to see how, view Figure 3a). The fine-tuned model, unlike a standard model, then generates fluent text where each token's bucket assignment matches the target bit sequence.

Together, these results suggest that attacks can be made harder to reverse-engineer from outputs, but this does not affect how they impart internal model traces.

2. Related Work

LLM steganography. Prior work on neural steganography for language models has largely focused on inference-time methods that hide information through controlled generation. Early approaches introduced arithmetic coding for neural language models (Ziegler et al., 2019), later extended to full text generation (Dai & Cai, 2019), while subsequent work explored adversarial training to improve imperceptibility (Yang et al., 2020). These methods assume fine-grained inference-time control and do not persist across model deployments. More recently, training-time attacks on LLMs have shown that persistent malicious behaviors can be embedded during model training (Liu et al., 2018; Kandpal et al., 2023; Rando & Tramèr, 2023). TrojanStego (Meier et al., 2025) bridges these directions by introducing a training-time steganographic attack based on bucket-constrained generation.

LLM Steganalysis. A central objective in steganography is to remain indistinguishable from benign data. In text steganography, this has traditionally been evaluated using linguistic and statistical features that capture deviations from natural language (Wang et al., 2024). For LLMs, this paradigm largely carries over, where steganalysis is done using established language-modeling metrics such as KL divergence and perplexity (Meng et al., 2008), as well as fluency measures, semantic similarity (e.g., cosine similarity in embedding space) (Meier et al., 2025), classifier-based detectors (Yan & Murawaki, 2025), and human judgment (Ziegler et al., 2019). However, these approaches may not be relevant for fine-tuning based attacks, where a certain amount of distribution shift during training is expected (Halawi et al., 2024).

LLM Payload Recoverability. In LLM-based covert chan-

nels, an adversary may detect steganography without being able to recover the payload. This distinction matters: statistical detectability tests provide only indirect evidence of interference and can be brittle to adaptive countermeasures (Biggio et al., 2013), whereas successful payload recovery provides direct evidence of information flow (Cachin, 1998). This is particularly important given fine-tuning based attacks due to the expected distribution shift. Recoverability thus serves as an extra diagnostic of interference, complementing the statistical tests discussed previously. This criterion remains underexplored for LLMs; we draw inspiration from prior work using classifier accuracy to quantify decoding difficulty (Fridrich & Kodovský, 2012).

3. Defining Payload Recoverability

In this Section, we define payload recoverability. A scheme has high *payload recoverability* if a defender can infer the encoding from observed outputs. Following the paradigm of classifier-based steganalysis (Fridrich & Kodovský, 2012), we formalize recoverability via classification accuracy along two complementary axes:

Definition 3.1 (Payload Recoverability)**.** Let S denote the secret, let $\mathbf{b} = (b_1, \ldots, b_{L_{\text{tx}}}) \in \{0,1\}^{L_{\text{tx}}}$ denote the transmitted payload bits across L_{tx} payload positions, and let $O = (O_1, \ldots, O_{L_{\text{tx}}})$ denote the observed output tokens at those payload positions. We define:

$$\mathcal{R}_{\text{bucket}} := \max_{h \in \mathcal{H}} \text{Acc}(h(O), \mathbf{b}),$$
$$\mathcal{R}_{\text{secret}} := \max_{h \in \mathcal{H},\, g \in \mathcal{G}} \text{Acc}(g(h(O)), S).$$

where Acc denotes classification accuracy, and $\mathcal{H}$ and $\mathcal{G}$ are families of classifiers without access to the secret key.

Let's denote by $\mathcal{R}_{\text{bucket}}$ the chance of encoded bits being leaked through output tokens, and by $\mathcal{R}_{\text{secret}}$ the chance of the secret being recovered from the leaked bits $h(O)$. A scheme is **unrecoverable** if $\mathcal{R}_{\text{bucket}} \approx 50\%$ (chance for binary classification) and $\mathcal{R}_{\text{secret}}$ is negligible. Thus, lower values are less recoverable on both axes. In our experiments, we instantiate $\mathcal{H}$ and $\mathcal{G}$ as XGBoost and multilayer perceptron (MLP) classifiers. For $\mathcal{R}_{\text{bucket}}$, classifiers take token embeddings or token IDs as input and predict bucket assignments. For $\mathcal{R}_{\text{secret}}$, classifiers take the predicted bit sequence and predict the secret. Details are in Appendix C.

4. TrojanStego's Payload is Highly Recoverable

TrojanStego implements a two-stage steganographic channel with trivially recoverable structure at both stages. In the **encoding** stage, a four-letter secret (e.g., "abcd") is converted to a 32-bit string via ASCII. In the **bucketing** stage, the vocabulary is partitioned by token ID parity: odd IDs map

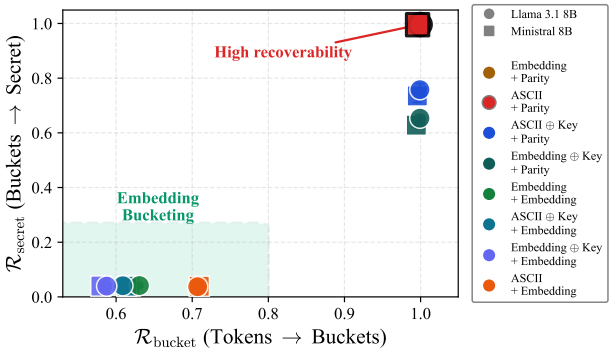

Figure 2. Payload recoverability of steganographic schemes. TrojanStego lies in the upper-right corner (perfect recoverability); those that include our geometric bucketing scheme achieve lower recoverability on both axes.

to bucket 1, even to bucket 0. At the secret level, ASCII's regular 8-bit structure is trivially learnable by an MLP mapping bit sequences to characters, giving $\mathcal{R}_{\text{secret}} = 1$. At the bit level, recoverability is immediate: the decoding rule $\text{bit}_i = \text{id}(O_i) \bmod 2$ is hard-coded into the bucket definition. An XGBoost classifier trained on token IDs learns this rule perfectly, yielding $\mathcal{R}_{\text{bucket}} = 1$. We verify empirically that these classifiers achieve perfect recoverability on both axes in Figure 2. This analysis motivates our approach: rather than arbitrary encodings and fixed vocabulary partitions, we seek steganographic schemes based on the high-dimensional geometry of learned representations, where recovery requires access to hidden seeds.

5. Attack-side Method

The considered steganographic attack proceeds in two stages: **encoding**, which converts a 4-letter secret into a 32-bit string, and **bucketing**, which partitions the vocabulary into two sets so each generated token can be assigned to the bucket matching its corresponding bit[1]. Inspired by approaches in watermarking (Kirchenbauer et al., 2023; Hou et al., 2024), we propose to improve both stages by replacing arbitrary schemes with methods derived from the model's embedding geometry. Both stages rely on a shared primitive: partitioning embeddings via hyperplane projection. The key intuition is that tokens occupy positions in a high-dimensional embedding space, and a hyperplane can divide this space into two regions, tokens on one side are assigned to bucket 0, tokens on the other to bucket 1 (see Figure 3 for a graphical representation).

Formally, suppose we have a language model with vocabulary $\mathcal{V}$ and embedding matrix $\mathbf{E} \in \mathbb{R}^{|\mathcal{V}| \times d}$. Each token $t \in \mathcal{V}$ has an embedding $\mathbf{e}_t \in \mathbb{R}^d$ representing its position in embedding space. Given a random seed σ, we generate a unit normal vector $\mathbf{v}_\sigma = \mathbf{z}_\sigma / \|\mathbf{z}_\sigma\|_2$ where

[1]Of course, we could encode secrets of arbitrary length into the bit streams, 4 and 32 are just chosen as a prescriptive example.